

FOURLINK - Version 4.0
Racing Systems Analysis - www.QUARTERjr.com

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Note: sample case for FOURLINK

General Data

Estimated 60 ft Time - sec 1.01
Maximum Acceleration - g's 2.75

Tire Rollout - inches 102.0

Static Weight Data - lbs %

Front Weight 1200 51.0
Rear Weight 1155 49.0

Total Weight 2355 100.0

Rear Suspension Data

Shock Mount Location - inches -5.00
Rear Spring Rate - lbs/inch 85

Shock Units: lbs per in/sec
Compression (Bump) Rate 66
Extension (Rebound) Rate 209

Wheelie Bar Length - inches 68.0

Center of Gravity Data

Wheelbase - inches 105.0
Horizontal CG - inches 53.6

Vertical CG - inches 17.0
Front Strut Lift - inches 2.0
Front Tire Lift - inches 1.0

Weight of Rear Axle Assembly 350

Four Link Geometry Data - Hole Code 2123

Link Bar	Axle End		Chassis End		Length	Angle	Forces and Components		
-----	x-in	y-in	x-in	y-in	inches	degrees	lbs	Horiz	Vertical
Upper	1.000	19.000	20.100	14.250	19.68	-14.0	-6391	-6244	1365
Lower	0.500	8.000	20.500	7.000	20.02	-2.9	11760	11757	-244
							Totals	5514	1121

Dynamic Weight Transfer

Front Weight - lbs 0
Rear Weight - lbs 2312
Wheelie Bar Force - lbs 43
Shock Separation - inches -0.6
Shock Damping Ratio 2.22

Instant Center Parameters

Horizontal IC - inches 56.5
Vertical IC - inches 5.2

Percent Anti-Squat - % 91
Initial Rear Tire Hit - lbs 683

Dynamic Chassis Analysis - Hole Code 2123

Time	Separation		Forces - lbs			Rear	Front	Wheelie
sec	in	in/sec	Spring	Shock	Mass	Tires	Tires	Bars
0.00	0.00	0.0	740	0	0	1155	1200	0
0.05	0.05	1.5	730	-636	42	1740	657	0
0.10	0.15	2.2	714	-933	39	2292	102	0
0.15	0.25	1.8	696	-755	-37	2234	84	0
0.20	0.33	1.2	684	-505	-44	2251	60	0
0.25	0.37	0.6	676	-255	-53	2273	29	0
0.30	0.39	0.0	674	-17	-39	2304	0	11
0.35	0.37	-0.5	677	61	0	2311	0	43
0.40	0.35	-0.5	681	61	0	2311	0	43
0.45	0.32	-0.5	685	61	0	2312	0	43
0.50	0.30	-0.5	689	62	0	2312	0	43
0.55	0.28	-0.5	693	62	-1	2311	0	43
0.60	0.25	-0.5	697	62	0	2312	0	43
0.65	0.22	-0.9	702	121	-92	2260	0	3
0.70	0.18	-0.9	710	118	9	2313	51	0
0.75	0.13	-0.8	717	101	9	2261	103	0
0.80	0.10	-0.7	723	93	10	2210	155	0

Average Rear Tire Force - lbs 2290

Rear Tire Force Variation - % 3.4

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Four Link Bar Geometry Data

Hole	Upper Link Bar					Lower Link Bar			
	Axle End		Chassis End			Axle End		Chassis End	
	x-in	y-in	x-in	y-in		x-in	y-in	x-in	y-in
1	1.000	20.000	20.100	14.250		0.500	9.000	20.500	9.000
2	1.000	19.000	20.100	13.250		0.500	8.000	20.500	8.000
3	1.000	18.000	20.100	12.250		0.500	7.000	20.500	7.000
4	1.000	17.000	20.100	11.250		0.500	6.000	20.500	6.000
5	1.000	16.000	20.100	10.250		0.500	5.000	20.500	5.000

Adjust Geometry: Axle Height - in 0.000 Chassis Height - in 0.000
Pinion Angle - deg 0.00

Calculated Four Link Bar Details - Display Limits

Shock Separation - in		Instant Center Locations - in
min -3.0		x-min 44.0 y-min 1.0
max 3.0		x-max 68.0 y-max 16.0

Percent Anti-Squat - %		Lower Link Bar Angle - degs
min 70		min -4.5
max 105		max -0.5

Calculated Four Link Bar Details**Display Limits: On**

Hole	Instant	Center	Shock	%Anti-	Tire		Hole	Instant	Center	Shock	%Anti-	Tire
Code	x-in	y-in	Sep.	Squat	Hit		Code	x-in	y-in	Sep.	Squat	Hit
5434	46.4	4.7	0.1	102	748		3112	62.7	5.9	-0.7	89	692
3223	51.5	5.5	0.0	100	775		3345	48.9	3.6	-0.8	88	567
4212	55.8	6.2	0.0	100	809		3234	56.5	4.2	-1.1	84	568
2234	48.9	4.6	-0.2	97	698		5445	51.5	3.5	-1.1	83	526
4445	44.9	3.8	-0.3	96	641		4223	62.7	4.9	-1.2	82	588
5323	55.8	5.2	-0.5	92	694		2245	52.9	3.4	-1.3	81	504
4334	51.5	4.5	-0.5	92	650		2134	61.5	3.9	-1.6	76	499
2123	56.5	5.2	-0.6	91	683		4345	56.5	3.2	-1.6	76	454
1245	47.2	3.7	-0.6	91	597		1145	56.9	3.2	-1.7	75	449
1134	52.9	4.4	-0.7	89	626		5334	62.7	3.9	-1.7	75	484